

Pure Periodicity for the Three-Move Subtraction Games $\{2, 5, c\}$

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

For an integer $c > 5$, $c \neq 7$, consider the subtraction game with moves $\{2, 5, c\}$ and its full Sprague–Grundy sequence, with the wall at zero. We prove the fixed-shape specialization of Manabe’s admissible-angle conjecture: the sequence is periodic from its first term if and only if $c \bmod 7 \in \{1, 2, 3, 5, 6\}$. In the positive cases its least period is $c + 2$ for residue 1, 7 for residues 2, 5, and $c + 5$ for residues 3, 6. The source criterion supplies these positive cases. For the two remaining residue classes we give exact ultimately periodic nim-value words. Their tails have period $c + 2$, but that period fails at zero; a transfer lemma then excludes every possible pure period. Thus the result concerns full nim values, not merely winning and losing positions.

1 Statement and context

Let $G_c(n)$ be defined for $n \in \mathbb{N}_0$ by

$$G_c(n) = \text{mex}\{G_c(n - s) : s \in \{2, 5, c\}, s \leq n\}, \quad (1)$$

where $\text{mex } \emptyset = 0$. We call G_c *purely periodic* if some $p > 0$ satisfies $G_c(n + p) = G_c(n)$ for every $n \geq 0$. This is the nim-value convention used here throughout; replacing $G_c(n)$ by the binary indicator of $G_c(n) = 0$ would be a different assertion.

Manabe attaches three admissible-angle sets to a primitive three-move shape and proves that membership produces a purely periodic game, including its least period [2, Theorem 32 and Proposition 46]. Necessity is posed there as Conjecture 50. For the shape $(a, b) = (2, 5)$, Lemma 14 and Theorems 28 and 57 of that paper specialize to

$$\Theta_7 = \{2, 5\}, \quad \Theta_{c+2} = \{1, 5\}, \quad \Theta_{c+5} = \{2, 3, 6\}, \quad (2)$$

with residues taken modulo 7. Zhang’s results determine other slices of the three-move problem [3]; they do not supply the all- c fixed-shape proof below. An informal anonymous note from 2009 states the same zero/nonzero preperiod classification for $\{2, 5, c\}$ without a proof and gives a discrepant positive-period table [1]. Accordingly, our contribution is the complete proof and the two exact negative-family words, not a claim that the classification was never stated.

Theorem 1. *For every integer $c > 5$ with $c \neq 7$, the sequence G_c is purely periodic if and only if*

$$c \bmod 7 \in \Theta_7 \cup \Theta_{c+2} \cup \Theta_{c+5} = \{1, 2, 3, 5, 6\}.$$

In the positive cases its least period is

$$\lambda(c) = \begin{cases} c+2, & c \equiv 1 \pmod{7}, \\ 7, & c \equiv 2, 5 \pmod{7}, \\ c+5, & c \equiv 3, 6 \pmod{7}. \end{cases}$$

There is no proper-divisor exception for this primitive non-harmonic shape.

The implication from membership in the union to pure periodicity, as well as the asserted least periods, follows at once from (2) and the results of Manabe just cited. This includes the endpoint $c = 6$. It remains to rule out residues 0 and 4; that is done at the full nim-value level in the next section.

2 Exact words for the non-admissible angles

For a finite word X , write X^m for m concatenated copies and X^ω for infinite repetition. Define

$$\begin{array}{lll} B = 0011021, & E = 0011223, & A = 0210210, \\ C = 0110210, & D = 011021, & F = 32203103, \\ H = 1001102, & I = 1021021, & J = 02. \end{array}$$

Lemma 2 (residue 4). *If $c = 7k + 4$ with $k \geq 1$, then*

$$(G_c(n))_{n \geq 0} = B^k E (A C^{k-1} D)^\omega. \quad (3)$$

The tail begins at $q = c + 3$ and the displayed tail period has length $c + 2$.

Lemma 3 (residue 0). *If $c = 7k$ with $k \geq 2$, then*

$$(G_c(n))_{n \geq 0} = B^k F (H^{k-1} I J)^\omega. \quad (4)$$

The tail begins at $q = c + 8$ and the displayed tail period has length $c + 2$.

Proof of Lemmas 2 and 3. Call the right side of either claimed identity $x = UW^\omega$, and put $q = |U|$ and $p = |W|$. Directly from the block lengths,

c	q	p
$7k + 4$	$7k + 7 = c + 3$	$7k + 6 = c + 2$
$7k$	$7k + 8 = c + 8$	$7k + 2 = c + 2$

For each n , form the padded recurrence signature

$$\sigma_c(n) = (x_{n-2}^*, x_{n-5}^*, x_{n-c}^*; x_n),$$

where a star means that the index is negative and that entry is omitted from the option set. Substitution of the displayed blocks gives exactly the option sets in Table 1. Each claimed value is their mex.

For completeness, we justify that this is a finite check uniform in k . Once $n \geq q + c$, all four entries in a signature lie in the periodic tail; the signature is therefore p -periodic. Inside that tail the third predecessor has cyclic offset

$$-c \equiv 2 \pmod{p}. \quad (5)$$

Table 1: Complete option-set/mex table for the two word families.

families	option set	mex	families	option set	mex
0, 4	$\emptyset$	0	0, 4	$\{0\}$	1
0, 4	$\{0, 2\}$	1	0, 4	$\{1\}$	0
0, 4	$\{1, 2\}$	0	0, 4	$\{1, 3\}$	0
0, 4	$\{0, 1\}$	2	0, 4	$\{0, 1, 3\}$	2
0, 4	$\{0, 1, 2\}$	3	0	$\{0, 3\}$	1
0	$\{0, 2, 3\}$	1	0	$\{1, 2, 3\}$	0

It consequently suffices to inspect $0 \leq n < q + c + p$. Split this interval at c , q , and $q + c$. Before q , the only variable-length part is a string of seven-letter B blocks; offsets -2 and -5 see one block and its adjacent boundaries, while the range $c \leq n < q$ has only three positions in the residue-4 family and eight in the residue-0 family. In the transition range $q \leq n < q + c$, the current length- c word and the aligned word of c -predecessors are

$$\begin{array}{c|c|c} c & (x_q, \dots, x_{q+c-1}) & (x_{q-c}, \dots, x_{q-1}) \\ \hline 7k+4 & AC^{k-1}D[0:4] & B[3:7]B^{k-1}E \\ 7k & H^{k-1}I & B[1:7]B^{k-2}F. \end{array} \quad (6)$$

Here $X[i:j]$ denotes the substring in zero-based positions $i, \dots, j-1$. Every repeated aligned pair in (6) has length seven. Adding another C/B pair in the first row or H/B pair in the second repeats an existing interior local factor and creates no new boundary. Finally, on the cyclic word W , the three offsets are $-2, -5, +2$ by (5); again, one copy of every block and each adjacent cyclic boundary exhausts the local factors. The zero-copy endpoint cases $k = 1$ and $k = 2$ are fixed checks. Expanding these finitely many interiors and boundaries is precisely Table 1.

Thus x_n satisfies (1) for every n . The wall recurrence determines its sequence uniquely by induction, so $x = G_c$ in both families. $\square$

The local expansion above can also be reproduced by the accompanying plain Python certificate. It records all ordered padded signatures (20 in residue 4 and 25 in residue 0), verifies their mex values exactly, and compares the formula with a direct recurrence for any requested finite parameter range. Those computations are checks of the finite expansion; the all- k step is the block-local argument in the proof.

3 Excluding every pure period

The following elementary observation is useful because merely showing that $c + 2$ is not a period from zero would not, by itself, exclude another pure period.

Lemma 4 (period transfer). *Let $(x_n)_{n \geq 0}$ be periodic from zero with period $d > 0$. If $p > 0$ is a period from some index q onward, then p is already a period from zero.*

Proof. Fix $n \geq 0$ and choose $m \geq 0$ so that $n + md \geq q$. Pure d -periodicity and tail p -periodicity give

$$x_{n+p} = x_{n+p+md} = x_{n+md} = x_n.$$

This holds for every n . $\square$

In both Lemmas 2 and 3, $p = c + 2$ is a tail period, but it is not a period from zero. Indeed,

$c \bmod 7$	$G_c(0)$	$G_c(c + 2)$
4	0	$E[6] = 3$
0	0	$F[2] = 2$.

If either sequence had any pure period d , Lemma 4 would force its tail period $c + 2$ to hold from zero, contradicting the table. Hence neither non-admissible family is purely periodic. Together with the positive cases furnished by (2), this proves Theorem 1.

Computational and generative-AI disclosure

OpenAI Codex assisted with mathematical exploration, construction and execution of the exact verification script, adversarial checking, and drafting and revision of this exposition. A separate model-context audit is not independent peer review. The finite computations served as exact sanity checks only; the universal conclusion rests on the block-local proof above. The author reviewed the manuscript and the supporting calculations and takes responsibility for all claims.

References

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